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A Project Report
on
ApnaDukaan - Empowering Local Retail through AI-Driven Commerce

submitted for partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

Computer Science and Engineering

by

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CERTIFICATE

This is to certify that the Project Report titled **"ApnaDukaan - Empowering Local Retail Retail through AI-Driven Commerce"** is being submitted by, **Ms. MURUGULA DIVYA**, **Ms. NEELISETTY HEMA SIRI, Ms. PERUNGULAM KANAKA LAYA SADHANA, Mr. MYLA SAI HARSHITH** bearing Registered Numbers **23BQ1A05E8, 23BQ1A05F8, 23BQ1A05H7, and 23BQ1A05F0** respectively of III B.Tech II semester Computer Science & Engineering is a record Bonafide work carried out by them. The results embodied in this report have not been submitted to any other University for the award of any degree.

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Submitted for Viva-voce Examination held on _____

Internal Examiner

External Examiner

DECLARATION

We, **Ms. MURUGULA DIVYA, Ms. NEELISETTY HEMA SIRI, Ms. PERUNGULAM KANAKA LAYA SADHANA, Mr. MYLA SAI HARSHITH**, hereby declare that the Project Report entitled "Deep Learning – Traffic Violation Detection" done by us under the guidance of **Dr.Sk.Khadar Zilani, Professor at Vasireddy Venkatadri Institute of Technology (A)** is submitted in partial fulfillment of the requirements for the award of Bachelor of Technology in Computer Science & Engineering. The results embodied in this report have not been submitted to any other university for the award of any degree**.**

DATE :

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SIGNATURE OF THE CANDIDATE(s)

1.

2.

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4.

ACKNOWLEGMENT

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INSTITUTE VISSION

To impart quality education through exploration and experimentation and generate socially conscious engineers, embedding ethics and values, for the advancement in Science and Technology.

INSTITUTE MISSION

- To educate students with practical approach to dovetail them to industry needs.
- To govern the institution with a proactive and professional management with passionate teaching faculty.
- To provide holistic and integrated education and achieve over all development of students imparting scientific and technical, social and cognitive, managerial and organizational skills.
- To compete with the best and be the most preferred institution of the studious and the scholarly.
- To forge strong relationships and linkage with the industry.

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Providing quality education to enable the generation of socially conscious software engineers who can contribute to the advancement in the field of computer science and engineering.

DEPARTMENT MISSION

- To equip the graduates with the knowledge and skills required to enable them to be industry ready.
- To train socially responsible, disciplined engineers who work with good leadership skills and can contribute for nation building.
- To make our graduates proficient in cutting edge technologies through student centric teaching-learning process and empower them to contribute significantly to the software industry.
- To shape the department into a centre of academic and research excellence.

COURSE OUTCOMES

CO 1: Articulate problem statement. (K2)

CO 2: Apply technial knowledge. (K3)

CO 3: Acquire contemporaray tools & technologies. (K2)

CO 4: Communicate and present the entire SDLC. (K3)

CO 5: Perform the role of a team member or lead in SDLC. (K3)

Program Educational Objectives (PEOs)

The Programme Educational Objectives of the B.Tech in Computer Science Engineering programme are given below and are numbered from PEO1 to PEO4.

PEO1	To provide the graduates with solid foundation in computer science and engineering along with fundamentals of Mathematics and Sciences with a view to impart in them high quality technical skills like modelling, analyzing, designing, programming and implementation with global competence and helps the graduates for life-long learning.
PEO2	To prepare and motivate graduates with recent technological developments related to core subjects like programming, databases, design of compilers and Network Security aspects and future technologies so as to contribute effectively for Research & Development by participating in professional activities like publishing and seeking copy rights.
PEO3	To train graduates to choose a decent career option either in high degree of employability / Entrepreneur or, in higher education by empowering students with ethical administrative acumen, ability to handle critical situations and training to excel in competitive examinations.
PEO4	To train the graduates to have basic interpersonal skills and sense of social responsibility that paves them a way to become good team members and leaders.

Program Outcomes (POs)

The B.Tech CSE program has documented measurable outcomes that are based on the needs of the program's stakeholders. The program outcomes, which are derived from ABET criteria, were first drafted in the academic year 2009–10 and later revised in 2010–11. The program outcomes that the department presently adopts are as follows:

1	**Engineering knowledge**	Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
2	**Problem analysis**	Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural science and engineering sciences.
3	**Design/development of solutions**	Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal and environmental considerations.
4	**Conduct investigations of complex problems**	Use research based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5	**Modern tool usage**	Create, select and apply appropriate techniques, resources and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

6	**The Engineer and Society**	Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7	**Environment and Sustainability**	Understand the impact of the professional engineering solutions in the societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8	**Ethics**	Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9	**Individual and Team Work**	Function effectively as an individual and as a member or leader in diverse teams, and in multidisciplinary settings.
10	**Communication**	Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11	**Project Management and Finance**	Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12	**Lifelong Learning**	Recognize the need for, and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change.

Program Specific Outcomes (PSOs)

PSO-1	**Professional Skills:**	The ability to understand, analyze and develop computer programs in the areas related to algorithms, system software, multimedia, web design, big data analytics and networking for efficient design of computer-based systems of varying complexity.
PSO-2	**Successful Career and Entrepreneurship:**	The ability to employ modern computer languages, environments, and platforms in creating innovative career paths to be an entrepreneur and a zest for higher studies/employability in the field of Computer Science & Engineering.

CO – PO ARTICULATION MATRIX

Course Outcomes:

CO	Description
CO1	Articulate problem statement. (K2)
CO2	Apply technical knowledge. (K2)
CO3	Acquire contemporary tools & technologies. (K2)
CO4	Communicate and present the entire SDLC. (C3)
CO5	Perform the role of a team member or lead in SDLC. (K3)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	2	1			2	2	3		2			1	
CO2	2	2	3	2	2		2	1	2	2	2	1	3	1
CO3		1	1	1	3							1	1	
CO4	1								3	3			1	1
CO5		1						2	3	3	2		1	2

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ABSTRACT

ApnaDukaan - Empowering Local Retail through AI-Driven Commerce

Problem Statement :

Local shopkeepers are facing severe disadvantages and significant business losses due to their inability to establish an online presence. This challenge stems from a lack of resources and technical expertise to compete effectively against established e-commerce giants, leading to declining sales and reduced customer reach for neighborhood stores.

Solution :

ApnaDukaan provides a vital solution by offering a digital platform for local stores to compete online, offering customers the convenience of e-commerce combined with the trust of neighborhood shopping. The platform integrates a LangChain-powered chatbot for real-time customer assistance, providing dynamic store ratings, personalized recommendations, and helping users find the freshest products nearby, ensuring a fast, trusted, and interactive local shopping experience.

The platform is a **three-interfaced e-commerce system** managing the grocery supply chain.

- **Customer:** Browsing, secure checkout, and chatbot support.
- **Shopkeeper/Admin:** Comprehensive inventory, pricing, and order fulfillment management with analytics.
- **Delivery:** Optimized logistics, order details, and secure pickup/delivery verification.

Objectives :

- Equip local shopkeepers with a robust digital platform for a strong online presence.
- Significantly increase local store sales and broaden customer base via an accessible e-commerce channel.
- Provide customers a convenient, fast, and interactive shopping journey, blending e-commerce with local trust.
- Leverage advanced AI (LangChain chatbot) for real-time support, personalized recommendations, and efficient product discovery.
- Enable local businesses to effectively compete against e-commerce giants by overcoming technological barriers.

Conclusion :

Ultimately, ApnaDukaan delivers an AI-driven e-commerce solution that revitalizes local businesses by bridging the digital divide and connecting them directly with their community.

CHAPTER – 1

INTRODUCTION

In today's rapidly evolving digital world, technology has significantly transformed the way businesses operate and customers interact with services. The rise of e-commerce platforms has made online shopping more convenient, accessible, and efficient. However, this digital transformation has created major challenges for local shopkeepers and small-scale retailers, who often lack the resources and technical expertise required to establish an online presence.

Local businesses traditionally depend on physical customers and neighborhood trust to sustain their operations. With the increasing popularity of large-scale online platforms, customers are gradually shifting towards digital shopping. As a result, local shopkeepers are facing reduced customer footfall, limited visibility, and declining sales. Many small retailers are unable to compete with established e-commerce companies due to lack of digital tools and efficient delivery systems.

Another important issue is the absence of a unified platform that connects local stores with nearby customers. Customers often face difficulty in identifying which nearby shop has the required product or whether it is available. This lack of information leads to inconvenience and inefficient shopping experiences.

To address these challenges, the project **"ApnaDukaan – AI-Driven Local E-Commerce Platform"** is proposed. The system provides a digital platform that connects customers, shopkeepers, and delivery personnel through an integrated and user-friendly interface.

A key feature of the platform is the integration of an Artificial Intelligence-based chatbot powered by LangChain. This chatbot provides real-time assistance by answering queries, suggesting products, and helping users find suitable options, making the platform more interactive and intelligent.

The system is designed with three major modules: Customer, Shopkeeper/Admin, and Delivery. The customer module allows users to browse products and place orders, the shopkeeper module manages inventory and orders, and the delivery module ensures smooth logistics and secure delivery verification.

The implementation of ApnaDukaan utilizes modern web technologies such as React for frontend development, Node.js and Express for backend services, and MongoDB for database management. These technologies ensure that the system is scalable, efficient, and capable of handling real-time interactions

Overall, ApnaDukaan bridges the gap between traditional retail and modern e-commerce by providing a smart, accessible, and AI-powered solution that enhances local business growth and improves customer convenience.

CHAPTER – 2

AIM AND SCOPE

The project titled **"ApnaDukaan – AI-Driven Local E-Commerce Platform"** aims to develop an intelligent and user-friendly web-based system that enables local shopkeepers to establish a strong digital presence and compete effectively in today's fast-growing e-commerce environment. With the increasing shift of customers towards online shopping, small-scale retailers often struggle to sustain their business due to lack of technological support. This project is designed to bridge that gap by providing a unified digital platform for local commerce.

The primary objective of this system is to empower local shopkeepers by offering them a simple and efficient platform where they can register their stores, showcase products, update inventory, and manage customer orders without requiring advanced technical knowledge. This helps small businesses expand their reach beyond physical boundaries and attract more customers.

At the same time, the system focuses on improving customer convenience by allowing users to browse products from nearby stores, check availability in real-time, compare options, and place orders easily. This eliminates the need for physical visits and saves time while maintaining the trust associated with local shopping.

Another important aim of the project is to enhance user experience through Artificial Intelligence. The integration of a LangChain-powered chatbot enables customers to interact with the system in a smart and efficient way. The chatbot provides instant responses, suggests relevant products, and assists users in making better purchasing decisions, thereby improving overall engagement.

The system also aims to streamline the delivery process by incorporating a dedicated delivery module. This module ensures efficient order handling, real-time tracking, and secure delivery verification using methods such as OTP authentication. It improves reliability and builds trust among users.

The scope of the project includes:

- Development of a scalable web-based e-commerce platform for local businesses
- Enabling real-time product browsing, searching, and ordering
- Integration of AI-based chatbot for customer assistance
- Efficient inventory and order management system for shopkeepers
- Secure and optimized delivery system with tracking and verification

Overall, ApnaDukaan aims to create a balanced ecosystem where local retailers can grow digitally, customers can enjoy convenient shopping, and the entire process is supported by intelligent and efficient technology.

2.1 EXISTING SYSTEMS

At present, local businesses and customers rely on a combination of traditional and modern methods for buying and selling products. However, these systems are not fully efficient and have several drawbacks.

1. Traditional Offline Stores

Local shopkeepers mainly depend on direct customer visits for sales. Although this approach ensures personal interaction and trust, it has limitations such as:

- Limited reach restricted to nearby customers
- No online visibility or promotion
- Dependence on physical presence of customers
- Difficulty in managing large customer demand

2. Large E-Commerce Platforms

Popular platforms like Amazon and Flipkart provide advanced online shopping services. However, they are not suitable for small retailers due to:

- High competition from large sellers
- Commission fees and service charges
- Complex registration and management process
- Limited visibility for small vendors

3. Social Media and Informal Selling

Some shopkeepers use messaging apps and social media platforms to sell products. While this provides some level of digital presence, it lacks:

- Structured product listing and categorization
- Proper inventory and order management
- Secure payment and delivery systems

4. Limitations of Existing Systems

Despite multiple options, the current systems suffer from:

- Lack of a unified platform for local shops
- No intelligent recommendation or AI support
- Inefficient communication between customers and shopkeepers
- Poor delivery coordination and tracking

2.2 PROPOSED SYSTEM

To overcome the limitations of existing systems, **ApnaDukaan proposes a centralized AI-powered e-commerce platform** that connects local shopkeepers, customers, and delivery personnel through a single integrated system. The platform is designed to simplify online selling for small retailers while providing customers with a convenient and intelligent shopping experience.

1. Digital Store Creation for Shopkeepers

The system allows local shopkeepers to easily register and create their own digital store without requiring advanced technical knowledge. Shopkeepers can:

- Add and update product details such as name, price, and quantity
- Manage inventory in real-time
- View and process customer orders

This helps small businesses move from traditional offline selling to an organized digital platform.

2. Customer Shopping Interface

Customers can access the platform through a simple and user-friendly interface. They can:

- Browse products from nearby stores
- Search for specific items
- Check product availability instantly
- Add items to cart and place orders securely

This reduces the need for physical store visits and saves time.

3. AI Chatbot Integration (LangChain)

A key feature of the system is the integration of an AI-powered chatbot. This chatbot:

- Provides real-time assistance to users
- Suggests products based on user queries
- Helps customers find the best available options
- Answers questions related to products, prices, and availability

This makes the platform more interactive and improves the overall user experience.

4. Order Management System

Once a customer places an order:

- The shopkeeper receives order details instantly
- The system updates order status (placed, packed, out for delivery, delivered)
- Both customer and shopkeeper can track the order

This ensures smooth communication and efficient order handling.

5. Delivery and Verification Module

The delivery module is designed to ensure secure and reliable order fulfillment. It includes:

- Assignment of delivery personnel
- Real-time order tracking
- OTP-based delivery verification for security

This reduces errors and ensures safe delivery of products.

6. Integrated System Workflow

All modules (Customer, Shopkeeper, Delivery, AI Chatbot) are connected through a centralized backend system. This ensures:

- Real-time data synchronization
- Efficient communication between modules
- Fast and reliable system performance

7. Benefits of the Proposed System

The ApnaDukaan platform offers several advantages:

- Enables local shopkeepers to go digital
- Expands customer reach for small businesses
- Provides convenient and fast shopping experience
- Uses AI to enhance user interaction
- Ensures secure and efficient delivery system

The proposed system acts as a complete digital ecosystem for local commerce, combining e-commerce functionality with Artificial Intelligence to create a smart, efficient, and user-friendly platform.

CHAPTER – 3

CONCEPTS AND METHODS

3.1 PROBLEM DESCRIPTION

In the current digital era, e-commerce has become a dominant mode of buying and selling products. Large online platforms provide customers with convenience, variety, and quick delivery services. However, local shopkeepers and small retailers face significant challenges in adapting to this digital transformation. Most of them lack the technical knowledge, resources, and infrastructure required to establish an online presence.

Local businesses mainly depend on physical customers and traditional selling methods. This limits their reach to a small geographical area and reduces their ability to compete with large-scale e-commerce companies. As customers increasingly prefer online shopping, local stores experience reduced footfall and declining sales.

Another major problem is the absence of a centralized platform that connects local shopkeepers with nearby customers. Customers often face difficulty in identifying which store has the required product, whether it is available, or if it meets quality expectations. This leads to inconvenience and inefficient shopping experiences.

Additionally, there is a lack of intelligent systems that can assist customers in making better purchasing decisions. Most existing systems do not provide real-time support, personalized recommendations, or interactive assistance. This results in poor user engagement and reduced customer satisfaction.

Delivery management is also a challenge in local commerce. Without a proper system, tracking orders, assigning delivery personnel, and ensuring secure delivery becomes difficult. This can lead to delays, miscommunication, and lack of trust between customers and shopkeepers.

Therefore, the main problem addressed in this project is the lack of a unified, intelligent, and efficient digital platform that enables local businesses to operate online, connects them with customers, and provides seamless shopping and delivery experiences.

The **ApnaDukaan** system aims to solve these issues by creating an AI-powered e-commerce platform that supports local retailers, improves customer convenience, and ensures efficient order and delivery management.

3.2 PROPOSED SOLUTION

To address the challenges identified in the problem description, the ApnaDukaan system proposes a **comprehensive AI-driven e-commerce solution** that integrates multiple modules into a single platform.

The system allows local shopkeepers to create digital stores, upload product details, and manage inventory efficiently. Customers can browse products from nearby stores, place orders, and receive them at their doorstep. The platform ensures smooth interaction between all users through a centralized system.

A key component of the proposed solution is the integration of an AI-powered chatbot using LangChain. This chatbot provides real-time assistance, answers user queries, and suggests relevant products based on customer preferences. It enhances user engagement and simplifies the shopping process.

The system also includes an order management module that handles order placement, processing, and tracking. Once an order is placed, it is forwarded to the respective shopkeeper, who prepares the order for delivery. The delivery module ensures that orders are assigned to delivery personnel and delivered securely using OTP verification.

By combining these features, the system provides:

- A digital marketplace for local businesses
- Convenient and efficient shopping experience for customers
- Intelligent assistance using AI chatbot
- Reliable and secure delivery system

Thus, ApnaDukaan acts as a complete solution that bridges the gap between traditional retail and modern e-commerce.

3.2.1 DATASET

The dataset used in the ApnaDukaan system consists of information related to users, shops, products, orders, payments, and delivery details. It plays a crucial role in managing the overall functionality of the platform, including product listing, order processing, inventory management, and delivery tracking. The dataset enables efficient communication between different modules such as customer, shopkeeper, and delivery systems, ensuring smooth and reliable e-commerce operations.

1. User Dataset

This dataset stores information about all users in the system, including customers, shopkeepers, and delivery personnel.

Attributes:

- UserID (Primary Key)
- Email

- Password

This dataset is used for authentication and role-based access control within the platform.

2. Shop Dataset

This dataset contains details of registered shops on the platform.

Attributes:

- ShopID (Primary Key)
- ShopName
- Location
- Rating
- ShopkeeperID (Foreign Key → UserID)

It helps in linking shopkeepers with their respective stores and managing store-related operations.

3. Product Dataset

This dataset stores details of all products available in different shops.

Attributes:

- ProductID (Primary Key)
- ProductName
- Price
- CategoryID (Foreign Key)
- ShopID (Foreign Key)

This dataset enables customers to browse products and allows shopkeepers to manage their product listings.

4. Category Dataset

This dataset is used to classify products into different categories.

Attributes:

- CategoryID (Primary Key)
- CategoryName

It helps in organizing products and improving search functionality.

5. Inventory Dataset

This dataset manages the stock details of products.

Attributes:

- InventoryID (Primary Key)
- ProductID (Foreign Key)
- Quantity
- LastUpdated

It ensures real-time tracking of product availability.

6. Order Dataset

This dataset stores information about customer orders.

Attributes:

- OrderID (Primary Key)
- CustomerID (Foreign Key → UserID)
- OrderDate
- TotalAmount
- OrderStatus

This dataset is central to managing order processing and tracking.

7. OrderItem Dataset

This dataset stores details of individual items in each order.

Attributes:

- OrderItemID (Primary Key)
- OrderID (Foreign Key)
- ProductID (Foreign Key)
- Quantity
- Price

It allows multiple products to be associated with a single order.

8. Payment Dataset

This dataset manages payment-related information.

Attributes:

- PaymentID (Primary Key)
- OrderID (Foreign Key)
- PaymentMethod
- PaymentStatus
- TransactionID

It ensures secure and trackable payment processing.

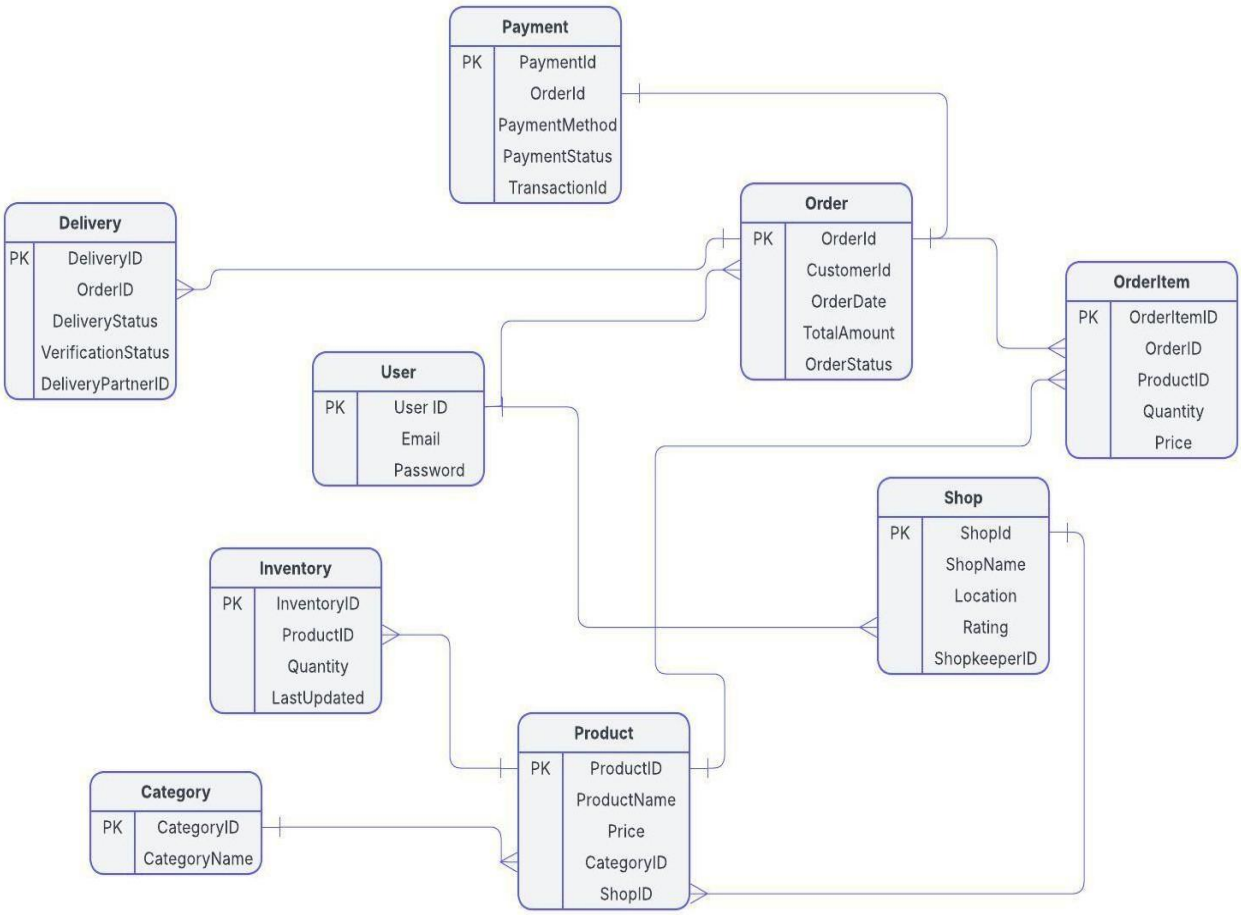
9. Delivery Dataset

This dataset handles delivery operations.

Attributes:

- DeliveryID (Primary Key)
- OrderID (Foreign Key)
- DeliveryStatus
- VerificationStatus
- DeliveryPartnerID (Foreign Key → UserID)

It ensures proper tracking and secure delivery verification (OTP-based).



3.2.2 DATA PREPARATION

Data preparation in the ApnaDukaan system plays an important role in ensuring efficient data handling and smooth functioning of the e-commerce platform. The system deals with structured data such as user details, product information, shop records, inventory, orders, payments, and delivery data. Proper preparation of this data helps in maintaining consistency, accuracy, and reliability across all modules.

The data preparation process includes collecting data from different sources such as user inputs, shopkeeper product entries, and system-generated order and payment records. The collected data is then organized into structured database tables based on the system design, ensuring proper relationships between entities like User, Shop, Product, and Order.

The dataset is cleaned by removing duplicate entries, correcting inconsistent values, and handling missing data fields. For example, product details such as name, price, and category are standardized to maintain uniformity. Similarly, user and order data are validated to ensure correctness and avoid errors during processing.

The data preparation process involves the following steps:

● **Data Structuring:** Organizing data into relational tables such as User, Product, Order, and Inventory for efficient storage and retrieval.

● **Data Cleaning:** Removing duplicate records, correcting invalid entries, and ensuring consistency in product and user data.

● **Data Validation:** Verifying input data such as product prices, quantities, and user details to maintain data integrity.

● **Data Integration:** Linking different datasets using relationships (e.g., Product linked to Shop, Order linked to User) to enable smooth system operations.

After preparation, the data is stored in a database such as MongoDB, which allows fast access and real-time updates. Proper data preparation ensures that the system performs efficiently in tasks like product search, order processing, inventory management, and delivery tracking.

By maintaining structured and clean data, the ApnaDukaan system achieves better performance, reduces errors, and provides a reliable experience for users, shopkeepers, and delivery personnel.

3.2.3 DATA PREPROCESSING

Data preprocessing in the ApnaDukaan system involves transforming and organizing the prepared data to ensure efficient system operations and accurate results. Since the platform handles structured data such as users, products, orders, and inventory, preprocessing focuses on data validation, formatting, and optimization for smooth execution of e-commerce functionalities.

Initially, the data stored in the database is accessed and processed based on user requests. Product data is filtered and sorted according to categories, price, and availability to provide relevant search results.

User inputs such as search queries, order details, and login credentials are validated before processing to prevent errors and ensure system security.

The preprocessing stage includes the following steps:

● **Data Validation:** Ensuring that all inputs such as product quantity, price, and user details are correct and within valid ranges.

● **Data Formatting:** Standardizing data formats such as dates, prices, and text fields for consistency across the system.

● **Filtering and Sorting:** Processing product data based on user preferences like category, price range, and availability.

● **Inventory Updates:** Automatically updating product quantities after order placement to maintain real-time stock information.

● **Order Processing:** Structuring order details and linking them with users, products, and payment data for efficient tracking.

The system processes data in real-time to ensure quick response and seamless interaction between modules. For example, when a user searches for a product, the system retrieves and filters relevant data instantly. Similarly, when an order is placed, the system updates inventory, generates order records, and initiates delivery processing.

Proper data preprocessing ensures that the system operates efficiently, reduces inconsistencies, and improves overall performance. It plays a crucial role in delivering accurate results, maintaining data integrity, and providing a smooth user experience in the ApnaDukaan platform.

3.2.4 DATA AUGMENTATION

Data augmentation in the ApnaDukaan system refers to enhancing and expanding the existing dataset to improve system performance, usability, and scalability. Unlike image-based systems, augmentation in this context focuses on enriching product data, user interactions, and system-generated information to provide better recommendations and smoother functionality.

The system improves dataset quality by continuously updating product information, adding new categories, and incorporating real-time data such as user preferences, order history, and shop ratings. This helps in making the platform more dynamic and responsive to user needs.

The following augmentation techniques are applied:

● **Product Data Expansion:** Adding new products, categories, and variations to increase the diversity of available items.

● **User Interaction Data:** Utilizing user search history, purchase patterns, and preferences to enhance recommendations.

● **Dynamic Updates:** Continuously updating inventory, pricing, and availability to reflect real-time changes.

● **Rating and Feedback Integration:** Including customer reviews and ratings to improve product visibility and decision-making.

● **Recommendation Enhancement:** Using collected data to provide personalized product suggestions through the AI chatbot.

These techniques help the system adapt to changing user behavior and market conditions, making it more efficient and user-friendly. Unlike static datasets, the ApnaDukaan system evolves over time by incorporating new data, which improves accuracy in search results and recommendations.

Proper data augmentation ensures that the platform remains scalable, responsive, and capable of handling real-world scenarios effectively. It enhances user experience and supports better decision-making for both customers and shopkeepers.

3.2.5 SYSTEM WORKING MECHANISM

The ApnaDukaan system follows a structured workflow that connects customers, shopkeepers, and delivery personnel through an integrated e-commerce platform. The system is designed to ensure smooth data flow, real-time updates, and efficient processing of user requests.

The working mechanism of the system is described in the following steps:

User Registration and Login

Users (customers, shopkeepers, and delivery personnel) register on the platform by providing basic details such as email and password. After successful registration, users can log in securely and access their respective modules

1. Customer Module

The customer module provides an interface for users to browse and purchase products easily.

- Customers register and log in to the platform securely.
- They can browse products from nearby shops and search based on categories or preferences.
- Customers can interact with the AI chatbot to get product suggestions and resolve queries.
- Products can be added to the cart and orders can be placed through a secure checkout process.
- Customers can track their orders in real-time and receive updates on order status.

This module focuses on providing a convenient and user-friendly shopping experience.

2. Shopkeeper Module

The shopkeeper module enables local retailers to manage their digital stores efficiently.

- Shopkeepers register and create their online shop on the platform.

- They can add, update, or remove products along with details such as price, category, and quantity.
- Shopkeepers receive order notifications from customers.
- They process orders by confirming availability and preparing items for delivery.
- Inventory is automatically updated after each order.

This module helps shopkeepers digitize their business and manage operations effectively.

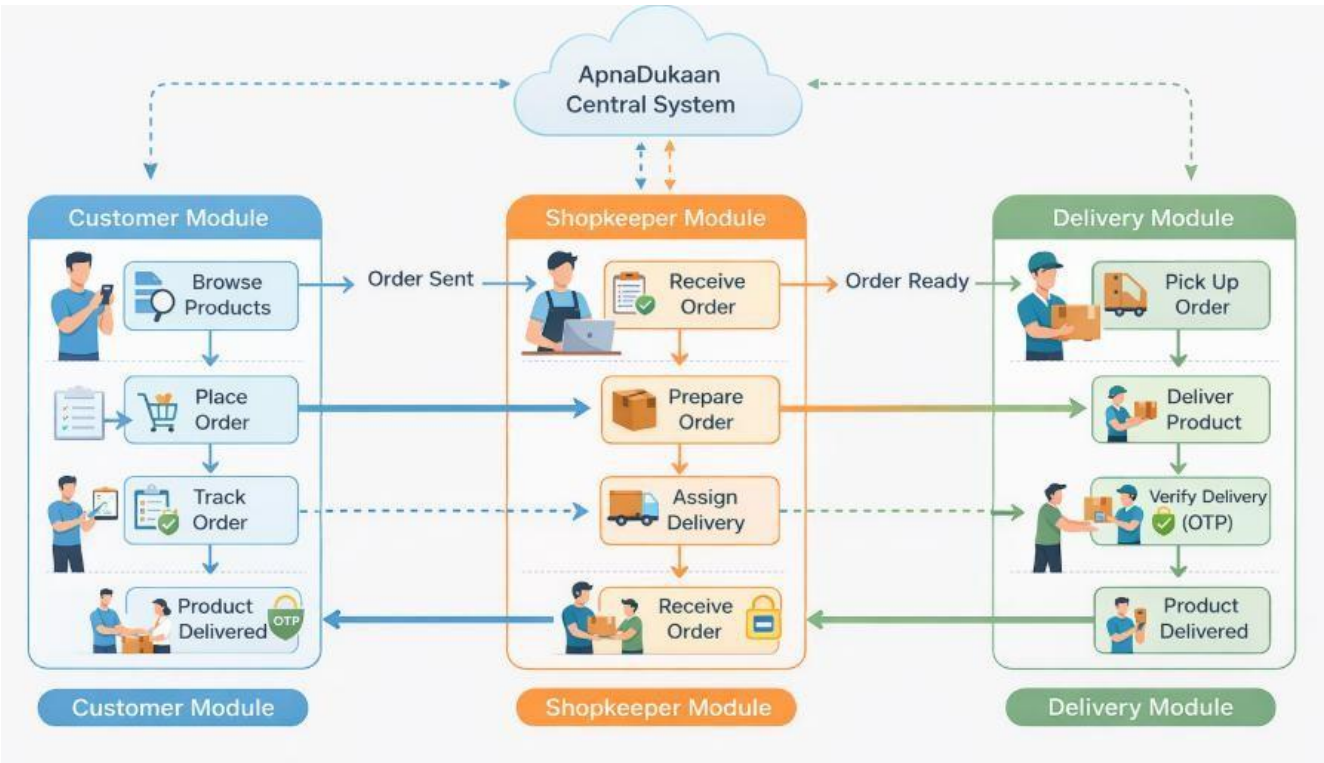
3. Delivery Module

The delivery module ensures efficient and secure delivery of orders to customers.

- Delivery personnel receive assigned orders from the system.
- They collect products from the respective shop and deliver them to customers.
- The system provides real-time tracking of delivery status.
- OTP-based verification is used at the time of delivery to ensure security.

This module ensures reliable and timely delivery of products.

The ApnaDukaan system integrates Customer, Shopkeeper, and Delivery modules into a unified platform, enabling smooth communication and real-time data flow. This modular approach improves efficiency, reliability, and overall user experience.



3.2.6 PERFORMANCE METRICS

Performance metrics are used to evaluate the efficiency, reliability, and effectiveness of the ApnaDukaan system. These metrics help in analyzing how well the platform performs in handling user requests, managing data, and providing a smooth shopping experience.

The following metrics are considered:

1. Response Time

Response time measures the time taken by the system to respond to user actions such as searching for products, placing orders, loading pages, and interacting with the chatbot. A faster response time improves user experience and ensures that the platform operates efficiently even during peak usage.

2. Accuracy

Accuracy refers to the correctness of system outputs, including product search results, order details, and chatbot responses. The system must provide relevant and precise information to users. High accuracy ensures better decision-making and increases user trust in the platform.

3. System Reliability

System reliability evaluates the ability of the platform to function continuously without failures, crashes, or data loss. A reliable system ensures that users can access services at any time and that transactions such as orders and payments are processed safely.

4. User Satisfaction

User satisfaction measures how effectively the system meets user expectations. It depends on factors such as ease of use, speed, accuracy, interface design, and overall experience. Feedback from users helps in improving system features and usability.

5. Scalability

Scalability determines the system’s ability to handle an increasing number of users, products, and transactions without affecting performance. A scalable system supports future growth and allows the platform to expand to multiple locations and larger user bases.

6. Data Consistency

Data consistency ensures that information such as product availability, order status, and inventory is updated correctly across all modules. This prevents errors such as incorrect stock levels or duplicate orders and maintains system integrity.

CHAPTER – 4

IMPLEMENTATION

4.1 TOOLS USED

The development of the ApnaDukaan system requires various tools and technologies to design, implement, deploy, and manage the platform effectively. These tools help in building the user interface, handling backend operations, managing data, and ensuring smooth system performance.

The major tools used in the implementation of the project are as follows:

1. Frontend Technologies

React JS
React JS is used to design and develop the user interface of the system. It helps in creating interactive and responsive web pages for customers, shopkeepers, and delivery personnel.

HTML (HyperText Markup Language)
HTML is used to structure the content of web pages such as forms, buttons, and product displays.

CSS (Cascading Style Sheets)
CSS is used to design the layout, colors, and styling of the application, making it visually appealing and user-friendly.

JavaScript
JavaScript is used to add dynamic behavior to the frontend, such as form validation, real-time updates, and user interactions.

2. Backend Technologies

Node.js
Node.js is used to build the backend of the application. It handles server-side logic, processes requests, and manages communication between frontend and database.

Express.js
Express.js is a framework used with Node.js to simplify backend development. It helps in creating APIs, handling routes, and managing server operations efficiently.

3. Database

MongoDB

MongoDB is used as the database to store and manage data such as users, products, orders, and delivery details. It provides fast data retrieval and supports scalability.

4. AI Technology

LangChain
LangChain is used to implement the AI chatbot in the system. It enables real-time interaction with users, provides product recommendations, and improves user experience.

5. Payment Integration

PayPal API
PayPal API is used to handle secure online transactions. It ensures safe payment processing and maintains transaction records.

6. Deployment Tools

Render
Render is used to deploy and host the backend services of the application. It provides a cloud platform for running server-side applications with automatic scaling and continuous deployment.

Vercel
Vercel is used to deploy the frontend of the application. It allows fast and efficient hosting of React applications with seamless integration and quick updates.

7. Development Tools

Visual Studio Code
VS Code is used as the development environment for writing, debugging, and testing the code.

Postman
Postman is used to test APIs and verify backend functionality.

8. Version Control

Git and GitHub
Git is used for version control to track changes in the code, and GitHub is used to store the project repository and enable team collaboration.

9. Web Browser

Web browsers such as Google Chrome are used to run and test the application, ensuring proper functionality and user experience.

4.2 CODE STRUCTURE / MODULES

The ApnaDukaan system is developed using a modular architecture, where different functionalities are divided into separate components. This modular design improves system maintainability, scalability, and ease of future enhancements. Each module performs a specific task and interacts with other modules through the backend system.

The main modules of the system are as follows:

1. User Authentication Module

This module handles user registration, login, and access control.

Functions:

- User registration (Customer, Shopkeeper, Delivery)
- Secure login and logout
- Password validation and authentication
- Role-based access control

This module ensures that only authorized users can access the system.

2. Customer Module

This module provides features for customers to browse and purchase products.

Functions:

- Browse products from nearby shops
- Search and filter products
- Add products to cart
- Place orders
- Track order status
- Interact with AI chatbot

This module focuses on providing a smooth and convenient shopping experience.

3. Shopkeeper Module

This module allows shopkeepers to manage their stores digitally.

Functions:

- Create and manage shop profile
- Add, update, and delete products
- Manage inventory
- Receive and process orders
- View sales and order details

This module helps shopkeepers efficiently manage their business operations.

4. Product Management Module

This module manages product-related data within the system.

Functions:

- Store product details (name, price, category, quantity)
- Update product availability
- Link products with shops and categories
- Handle product search queries

It ensures proper organization and accessibility of product data.

5. Order Management Module

This module handles the complete lifecycle of orders.

Functions:

- Create order records
- Update order status (placed, packed, delivered)
- Link orders with customers and products
- Manage order history

This module ensures smooth order processing and tracking.

6. Payment Module

This module manages payment transactions securely.

Functions:

- Process online payments using PayPal
- Store transaction details
- Update payment status
- Link payments with orders

It ensures secure and reliable financial transactions.

7. Delivery Module

This module manages delivery operations.

Functions:

- Assign orders to delivery personnel
- Track delivery status

- Provide real-time updates
- Verify delivery using OTP

This module ensures safe and efficient delivery of products.

8. AI Chatbot Module

This module provides intelligent assistance to users.

Functions:

- Answer user queries
- Suggest products
- Assist in product search
- Provide real-time interaction

It enhances user experience and makes the system interactive.

9. Database Module

This module handles data storage and retrieval.

Functions:

- Store user, product, order, and payment data
- Manage relationships between entities
- Ensure data consistency and security

This module acts as the backbone of the system.

4.3 RESULTS AND OUTPUT SCREENSHOTS

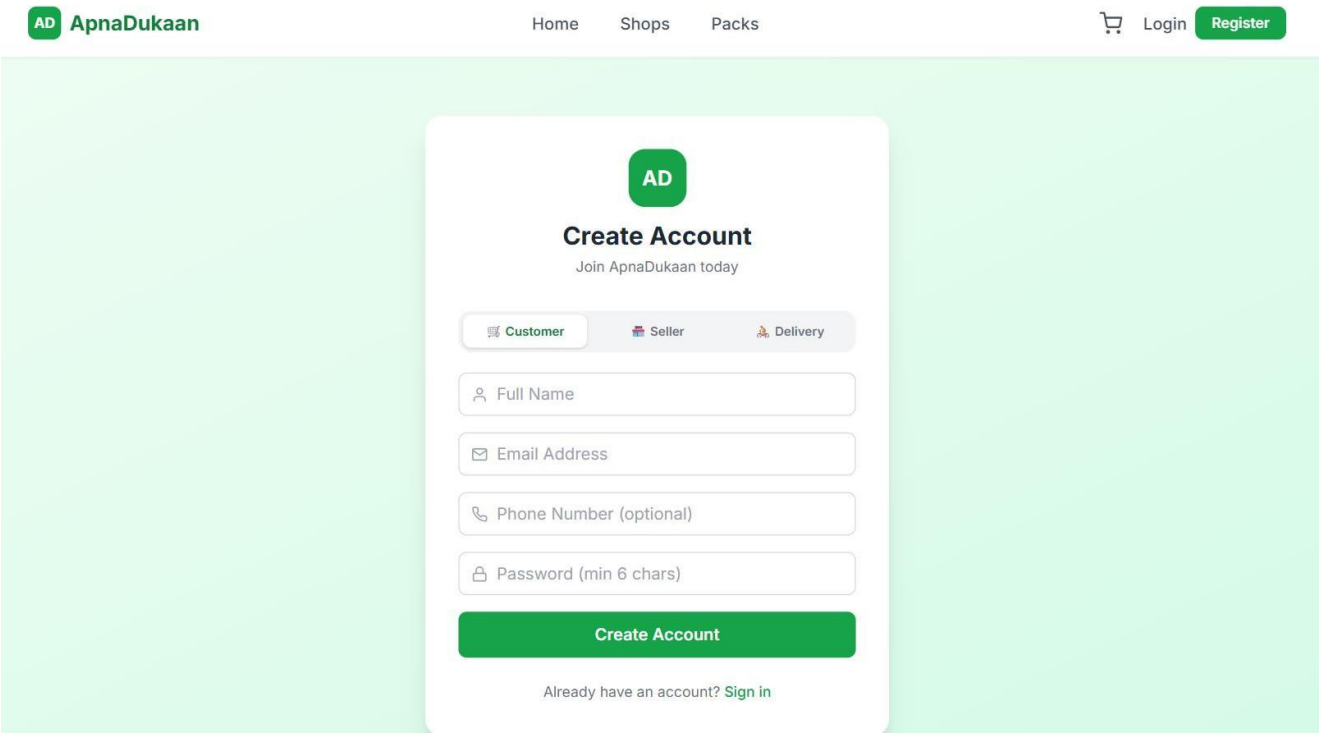
After the successful implementation of the ApnaDukaan system, the platform is able to perform all the core functionalities effectively. The system provides a seamless interaction between customers, shopkeepers, and delivery personnel, ensuring smooth e-commerce operations.

The results demonstrate that the application works efficiently in handling product browsing, order processing, payment transactions, and delivery tracking. The integration of the AI chatbot also enhances user interaction by providing real-time assistance and product recommendations.

The following outputs are observed from the system:

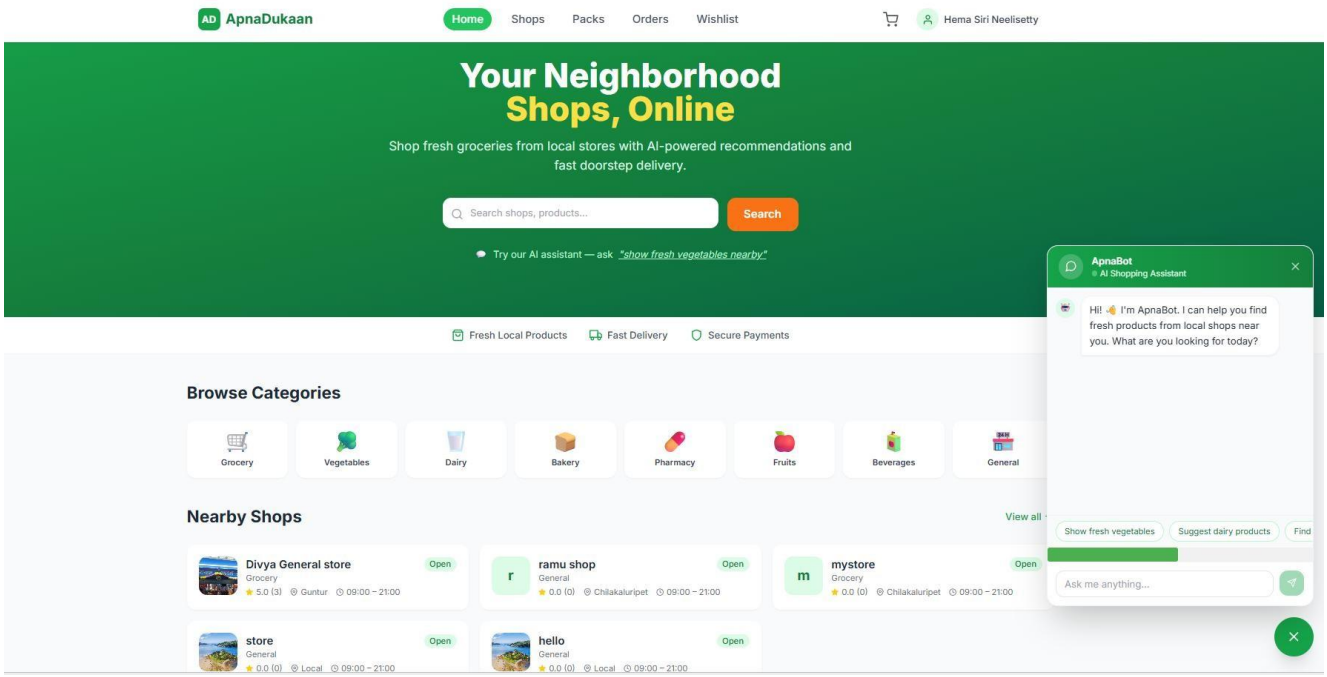
1. User Registration and Login

Users can successfully create accounts and log in securely to access the platform.



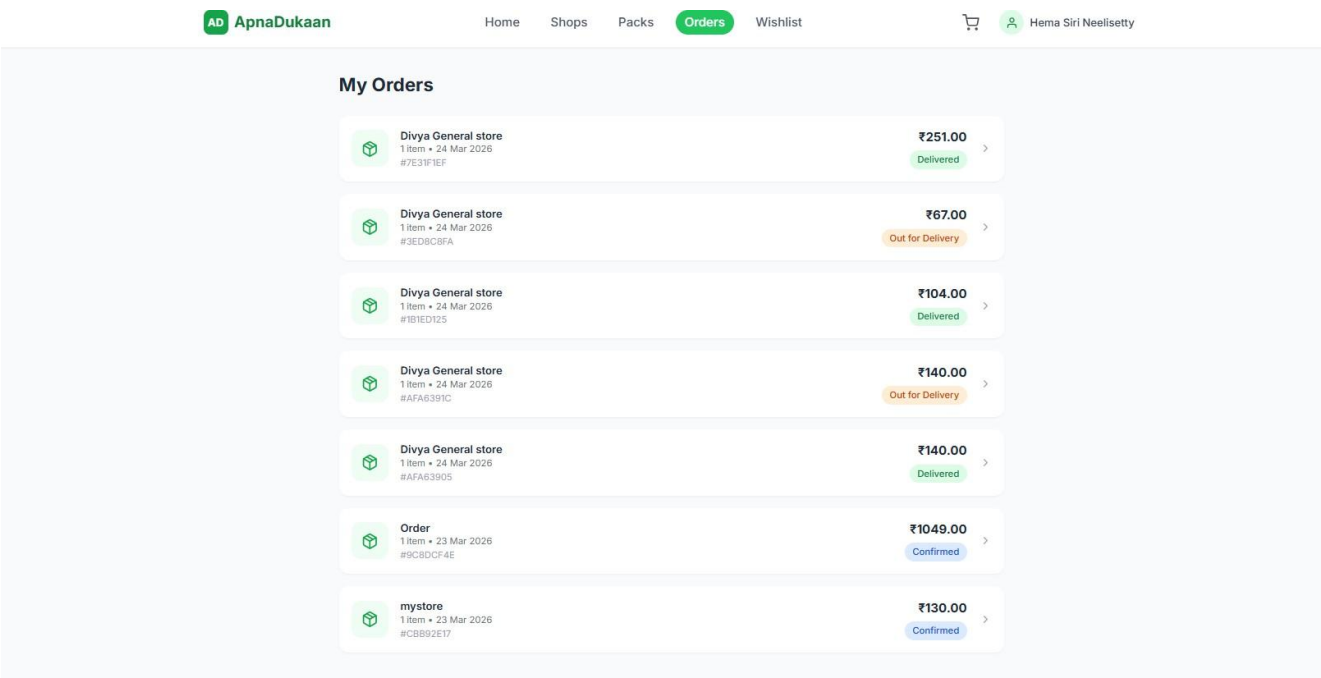
2. Product Browsing Interface & AI Chatbot

Customers can view products from nearby shops, search items, and filter based on categories and availability. The chatbot provides real-time assistance, answers queries, and suggests products.



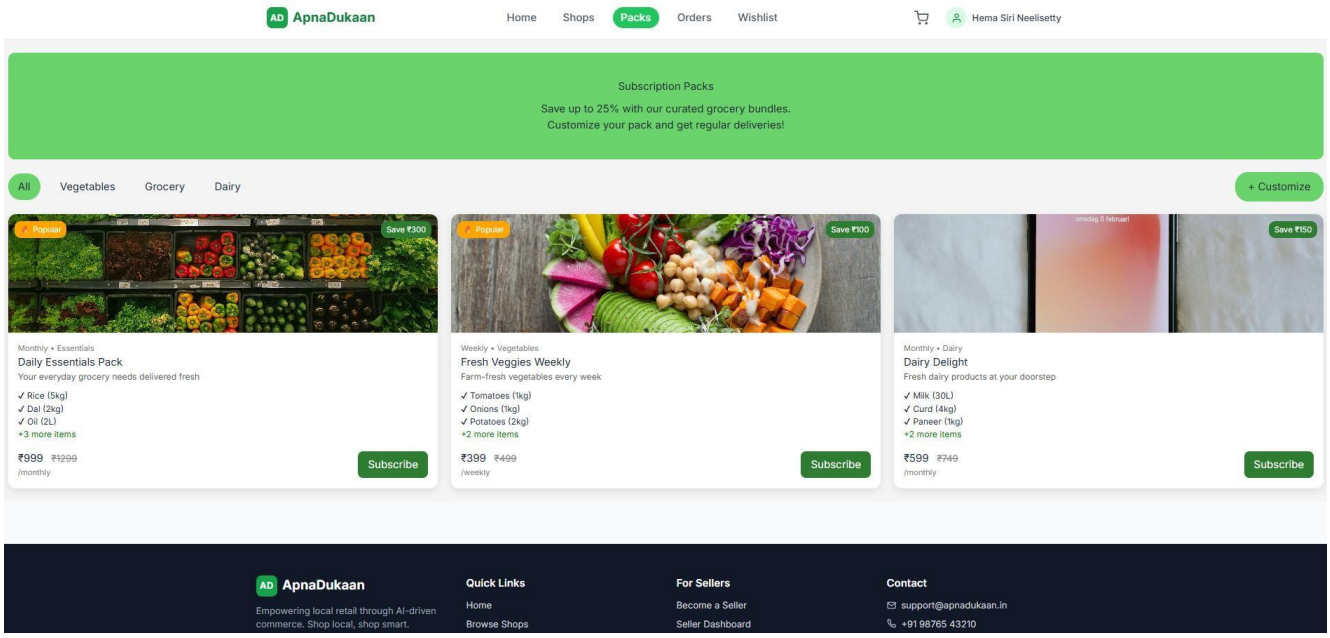
3. Shopping Cart and Order Placement

Customers can add products to the cart and place orders through a secure checkout process.



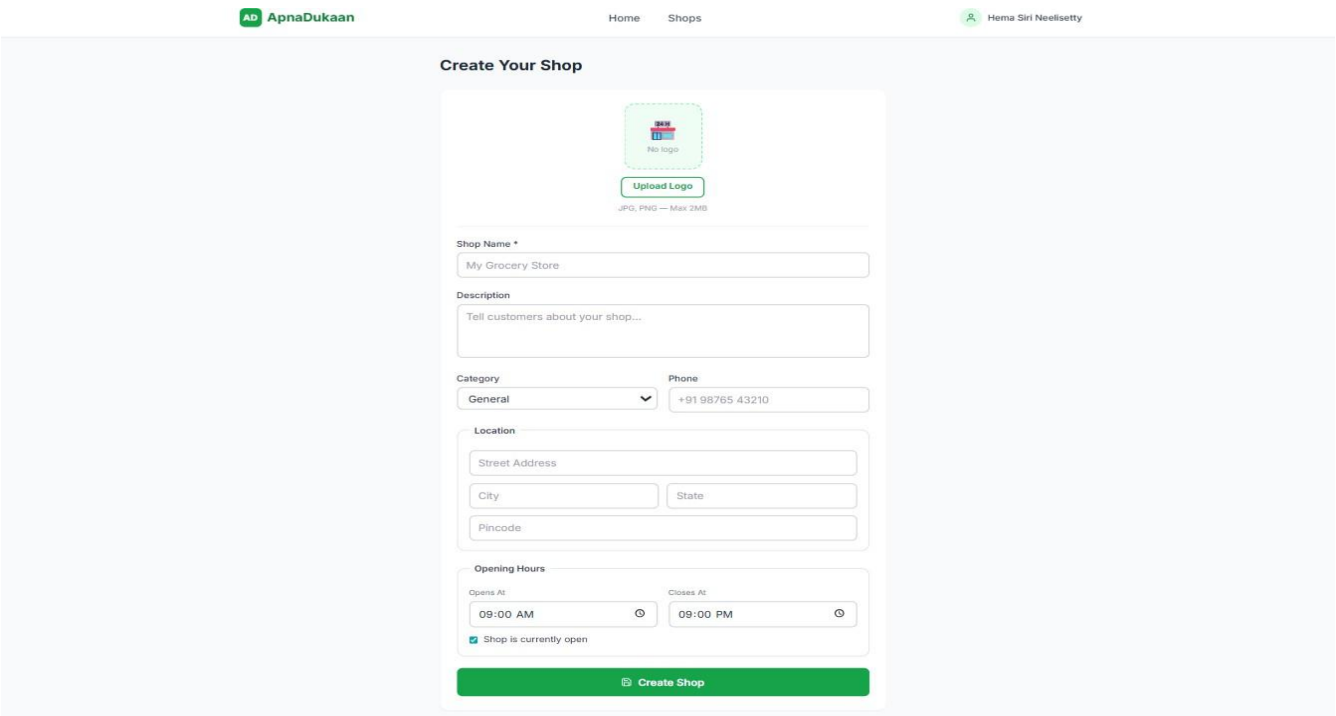
4. Subscription Packs Page

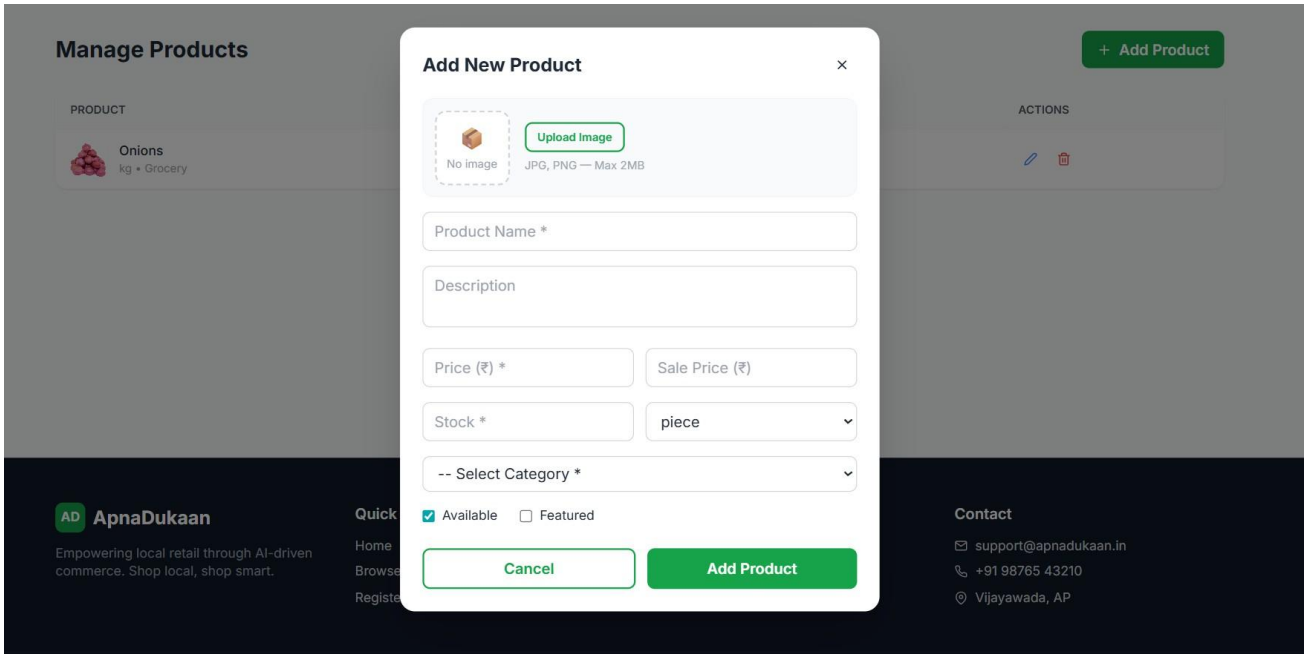
Each pack displays product details, pricing, and savings, allowing users to choose suitable options. This feature provides convenience by enabling regular deliveries and cost-effective shopping.



5. Shopkeeper Dashboard

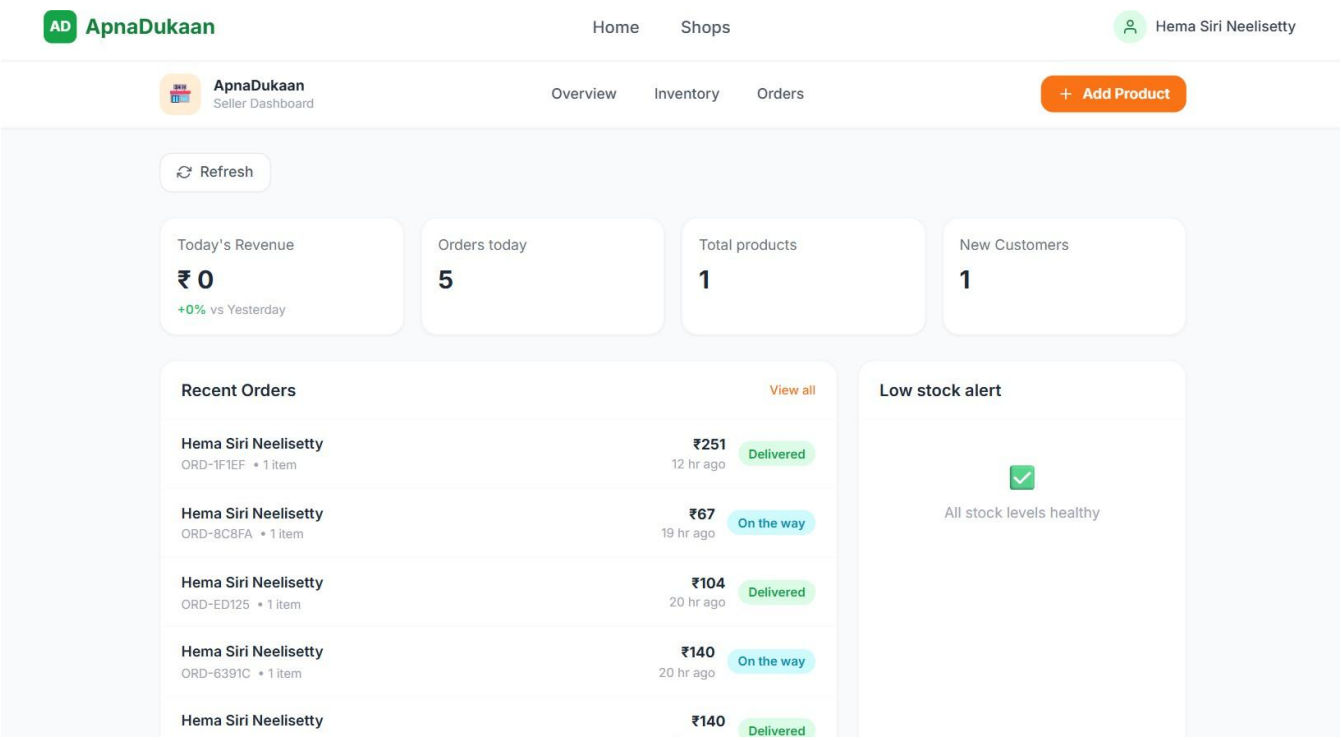
Shopkeepers can manage products, update inventory, and view incoming orders.

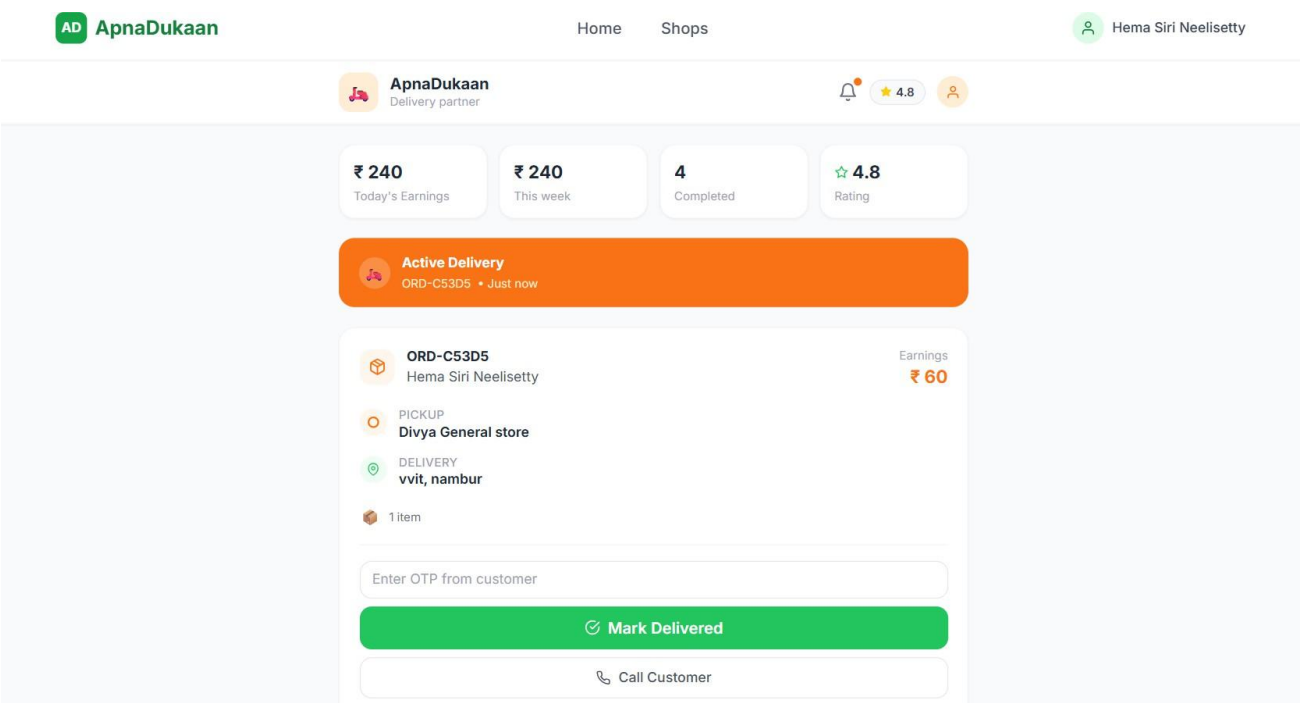




6. Delivery Module

Delivery personnel receive order details and complete deliveries with OTP verification.





CHAPTER – 5

CONCLUSION AND SCOPE

CONCLUSION

The ApnaDukaan system is developed to provide an effective digital solution for local shopkeepers, enabling them to establish an online presence and compete in the rapidly growing e-commerce environment. In today's technology-driven world, many small-scale retailers face difficulties due to lack of digital tools and limited customer reach. This project successfully addresses these challenges by creating a centralized platform that connects customers, shopkeepers, and delivery personnel.

The system allows shopkeepers to manage their products, inventory, and orders efficiently, while customers can conveniently browse items, place orders, and receive deliveries at their doorstep. The platform ensures smooth communication between all modules and provides real-time updates, improving overall system efficiency. The integration of an AI-powered chatbot further enhances user experience by offering instant assistance, product suggestions, and better interaction.

Additionally, the system incorporates secure payment processing and OTP-based delivery verification, ensuring reliability and trust in transactions. The modular architecture of the system makes it scalable, maintainable, and adaptable for future enhancements.

Overall, the ApnaDukaan platform successfully bridges the gap between traditional retail and modern e-commerce. It empowers local businesses, improves customer convenience, and provides a smart, efficient, and user-friendly solution for digital commerce.

FUTURE SCOPE

Although the ApnaDukaan system provides an effective solution for local e-commerce, there are several opportunities for further enhancement and expansion. One major improvement is the development of a mobile application, which would allow users to access the platform anytime and improve overall usability and reach.

The system can be enhanced by integrating advanced Artificial Intelligence features such as voice-based search, personalized product recommendations, and smarter chatbot capabilities. These features can improve user interaction and make the shopping experience more efficient and engaging.

Another important enhancement is the implementation of real-time analytics and demand prediction, which can help shopkeepers manage inventory more effectively and reduce product wastage. Integration with map-based tracking systems can provide accurate delivery tracking and improve logistics management.

The platform can also be expanded to support multiple cities and a larger user base, making it more scalable. Additionally, integrating multiple payment gateways, improving security features, and adding rating and review systems can further enhance reliability and user trust.

n addition, future versions of the system can include features such as online payment gateways with multiple options, real-time delivery tracking with maps, and enhanced security mechanisms. Integration with external platforms and APIs can further improve functionality and user experience.

With these improvements, ApnaDukaan has the potential to evolve into a comprehensive, intelligent, and scalable e-commerce solution that significantly contributes to the growth of local businesses and digital commerce.

REFERENCES

[1] M. Jason, "Building Scalable Web Applications with MERN Stack," IEEE Software, vol. 40, no. 2, pp. 45–52, 2023.

[2] S. Gupta and R. Sharma, "Design and Implementation of E-Commerce Systems," International Journal of Computer Applications, vol. 182, no. 10, pp. 1–6, 2022.

[3] A. Patel et al., "AI-Based Recommendation Systems in E-Commerce Applications," IEEE Access, vol. 11, pp. 10234–10245, 2023

[4] G. Adomavicius and A. Tuzhilin, "Toward the next generation of recommender systems: A survey of the state-of-the-art," *IEEE Transactions on Knowledge and Data Engineering*, vol. 17, no. 6, pp. 734–749, 2005.

[5] S. K. Singh and R. K. Sharma, "E-commerce recommendation system using machine learning algorithms," *International Journal of Advanced Computer Science and Applications*, vol. 11, no. 6, pp. 210–216, 2020.

[6] A. K. Jain and B. B. Gupta, "A survey on recommendation systems for e-commerce applications," *Journal of Network and Computer Applications*, vol. 178, p. 102940, 2021.

[7] S. Sarker, "Digital transformation and local business growth: A study on small-scale retailers," *Journal of Business Research*, vol. 135, pp. 320–330, 2021.